

Olga Isupova

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Education

- 2014–2017** *PhD*
The University of Sheffield, Department of [Automatic Control and Systems Engineering](#), UK
Thesis: *Machine learning methods for behaviour analysis and anomaly detection in video*.
Methodology included topic modeling, Bayesian nonparametrics, change point detection, approximated and sampling inference schemes, image and video processing
Supervisor: [Prof. Lyudmila Mihaylova](#), Professor of Signal Processing and Control.
- 2010–2013** Finished [Yandex School of Data Analysis](#), Russia
Master level program of [Data Analysis](#)
Study on algorithms and data structures, probabilistic methods in decision making theory, machine learning theory, convex theory and optimisation, linguistics and machine translation and others
- 2007–2012** *Specialist in Mathematics and System Programming* with Honours, degree equals to Masters
Lomonosov Moscow State University, Department of [Mathematical Methods of Forecasting](#), Russia
Thesis: *Hierarchical topic modeling in document collections with probabilistic latent semantic analysis*
Supervisor: [Prof. Konstantin Vorontsov](#), Professor of Russian Academy of Sciences

Employment History

- 12.2017 - present** *Postdoctoral Research Assistant in Machine Learning*
University of Oxford
Machine learning for disaster management and environment protection: i) development of joint learning methods for consolidated answers from crowdsourced labels and convolutional neural networks; ii) object detection on satellite imagery: [elephants](#), schools and tailing dams with noisy and not always reliable labels; iii) inference of socio-economic impact of droughts
- 04.2017 - 11.2017** *KTP Research Associate*
The University of Liverpool
Knowledge transfer partnership [project](#) with [Fletchers Solicitors](#). Development of algorithms predicting success of legal cases based on structured data, free text clients' stories and evidence records
- 04.2014 - 03.2017** *Marie Curie Early Stage Researcher in Marie Curie FP7 ITN TRAX project*
The University of Sheffield
"Knowledge Extraction from data" research topic. Development of new algorithms for information extraction from complex, heterogeneous data. Main applications: Bayesian methods for anomaly detection and sparse modelling. Research visits to [Automatic Control Group](#) at [the Linköping University](#), Sweden. Joint work with [Prof. Fredrik Gustafsson](#) on change point detection with Gaussian processes.
- 09.2011 - 02.2014** *Research Associate*
Croc Inc
Machine learning research and development: design of data mining and outlier filtering algorithms; data pre-processing
Business application functional implementation: design of business processes for data analysis, automation of developed algorithms, customer interviewing
- 06.2011 - 09.2011** *Analyst of the Customer Relationship Management department*
Tehnosila – electrical retailer
Data analysis in retail operations: sampling customers for target marketing campaigns; analysis of communication channels, results of marketing actions; statistical analysis of customer databases

Expertise

■ Maths

Probability theory, Statistics, Stochastic processes
Machine learning, Bayesian nonparametrics, Deep learning
Graphical models, Approximate Inference, MCMC

■ Programming and Miscellaneous

C/C++, Matlab, Python
Tensorflow, Keras
Databases, SQL

Other research related activity

■ Teaching

- Revision class for C24 Advanced Probability course, Department of Engineering Science, University of Oxford, Trinity term 2019
- Laboratory assistant and task designer for the master course “Intelligent and Vision Systems”, Department of Automatic Control and Systems Engineering, the University of Sheffield, Spring term 2016
- Assessing the entering exam and exam for the undergraduate course “Mathematical Foundation of Prediction Theory”, Department of Mathematical Methods for Forecasting, Lomonosov Moscow State University, Spring 2013

■ Talks

- “Dynamic topic models for video analysis and scene understanding” at the University of Surrey, 2019
- “Deep learning with noisy crowdsourced labels: Applications in disaster management” at 8th Oxford-Man Institute Machine Learning Workshop, 2019
- “Discovering dynamic topic transitions in topic models” at the University of Oxford Computational Statistics and Machine Learning Seminar, 2019
- “Modern video analytics systems” at Sheffield Festival of Science and Engineering as part of British Science week, 2016
- “Abnormal behaviour detection in video” at the PGR symposium at the University of Sheffield, 2014

■ Served as chairs and organisers

- Program committee of AAAI Conference on Artificial Intelligence, 2020
- Program committee of NeurIPS Machine learning for the developing world workshop, 2019
- Program committee of NeurIPS All of Bayesian nonparameterics workshop, 2018
- Session chair at the 19th International Conference on Information Fusion, 2016
- Organising committee of the summer school for the TRAX project, Sheffield, 2015
- Athena Swan Committee at the department of ACSE, the University of Sheffield, 2014–2017

■ Served as reviewers

- IEEE Transactions of Neural Networks and Learning Systems
- Signal Processing Elsevier journal
- Conference on Neural Information Processing Systems (NeurIPS) **top 400 reviewer at NeurIPS 2019**
- International Conference on Learning Representations (ICLR)
- International Conference on Machine Learning (ICML) **top 5% reviewer at ICML 2019**
- American Control Conference
- IEEE International Conference on Multisensor Fusion and Integration for Intelligent Systems (MFI)

Awards

- **Best paper award at NeurIPS 2018 Workshop on Machine Learning for the Developing World**

- **Springer Thesis Series for Outstanding Ph.D. Research**

PhD thesis is selected for publishing in the [Springer Thesis Series for Outstanding Ph.D. Research](#)

- **IET Control and Automation Doctoral Dissertation Awards 2017**

The second place in the [Institution of Engineering and Technology's \(IET\) Control and Automation Doctoral Dissertation Awards 2017](#)

Publications

Journals

O. Isupova, D. Kuzin, L. Mihaylova. “*Learning Methods for Dynamic Topic Modeling in Automated Behaviour Analysis*”, in IEEE Transactions on Neural Networks and Learning Systems, vol. 29, no. 9, 2018

D. Kuzin, **O. Isupova**, L. Mihaylova. “*Spatio-Temporal Structured Sparse Regression with Hierarchical Gaussian Process Priors*”, in IEEE Transactions on Signal Processing, vol. 66, no. 17, 2018

O. Isupova, D. Kuzin, F. Gustafsson, L. Mihaylova. “*Change Point Detection with Gaussian Processes*”, in IEEE Transactions on Pattern Analysis and Machine Intelligence, under review, 2019

Z. Li, L. Mihaylova, **O. Isupova**, L. Rossi. “*Autonomous Flame Detection in Videos With a Dirichlet Process Gaussian Mixture Color Model*”, in IEEE Transactions on Industrial Informatics, vol. 14, no. 3, 2018

Conferences

D.Kuzin, **O. Isupova**, L. Mihaylova. “*Baysian neural networks for sparse coding*”, in Proceedings of the International Conference on Acoustics, Speech, and Signal Processing, 2019, Brighton, UK

D.Kuzin, L. Yang, **O. Isupova**, L. Mihaylova. “*Ensemble Kalman Filtering for Online Gaussian Process Regression and Learning*”, in Proceedings of the 21st International Conference on Information Fusion, 2018, Cambridge, UK

R.Chen, M.Hawes, **O. Isupova**, L. Mihaylova, H.Zhu. “*Online Vehicle Logo Recognition Using Cauchy Prior Logistic Regression*”, in Proceedings of the 20th International Conference on Information Fusion, 2017, Xi'an, China

D.Kuzin, **O. Isupova**, L. Mihaylova. “*Structured Sparse Modelling with Hierarchical GP*”, in Proceedings of the Signal Processing with Adaptive Sparse Structured Representations Workshop, 2017, Lisbon, Portugal. Abstract

Z.Li, **O. Isupova**, L. Mihaylova, L.Rossi. “*Autonomous Flame Detection in Video Based on Saliency Analysis and Optical Flow*”, in Proceedings of the IEEE International Conference on Multisensor Fusion and Integration for Intelligent Systems, 2016, Baden-Baden, Germany

O. Isupova, D. Kuzin, L. Mihaylova. “*Dynamic Hierarchical Dirichlet Process for Abnormal Behaviour Detection in Video*”, in Proceedings of the 19th International Conference on Information Fusion, 2016, Heidelberg, Germany

D. Kuzin, **O. Isupova**, L. Mihaylova. “*Compressive Sensing Approaches for Autonomous Object Detection in Video Sequences*”, in Proceedings of the 10th Workshop Sensor Data Fusion: Trends, Solutions, and Applications, 2015, Bonn, Germany

O. Isupova, L. Mihaylova, D. Kuzin, G. Markarian, F. Septier. “*An Expectation Maximisation Algorithm for Behaviour Analysis in Video*”, in Proceedings of the 18th International Conference on Information Fusion, 2015, Washington, USA

O. Isupova, D. Kuzin, L. Mihaylova. “*Abnormal Behaviour Detection in Video Using Topic Modeling*”, in Proceedings of the University of Sheffield Engineering Symposium, 2015, Sheffield, UK. Abstract

O. Isupova. “*Matching a Hierarchical Topic Model with External Subject Authorities*”, in Proceedings of the XIX International Conference of Students and Young Scientists “Lomonosov-2012” 2012, Moscow, Russia. Abstract

Workshops

O. Isupova, D. Kuzin, B. Simmons, S. Reece. “*Rapid Post Disaster Damage Mapping with Satellite Imagery, Citizen Scientists and Machine Learning*”, UK Alliance for Disaster Research Annual Conference, 2019, Newcastle, UK

O. Isupova, D. Kuzin, B. Simmons, S. Reece. “*Disaster response: Leveraging wide area earth observation with machine learning*”, General Assembly of the European Geosciences Union, 2019, Vienna, Austria

O. Isupova, Y. Li, D. Kuzin, S. J Roberts, K. Willis, S. Reece. “*BCCNet: Bayesian Classifier Combination Neural Network*” (**Best paper award**), NeurIPS 2018 Workshop on Machine Learning for the Developing World, 2018, Montreal, Canada

O. Isupova, D. Kuzin, L. Mihaylova. “*Discovering Dynamic Topic Transitions in Topic Models*”, NeurIPS 2018 All of Bayesian Nonparametrics Workshop, 2018, Montreal, Canada

D. Kuzin, **O. Isupova**, L. Mihaylova. “*Uncertainty Propagation in Neural Networks for Sparse Coding*”, NeurIPS 2018 Bayesian Deep Learning Workshop, 2018, Montreal, Canada

O. Isupova, D. Kuzin, L. Mihaylova. “*Anomaly Detection in Video with Bayesian nonparametrics*”, ICML 2016 Anomaly Detection Workshop, 2016, New York, USA